Run report: Cheon P-224-shape scaling study, run 2026-10-22T0538Z

Descriptive record of one measured run (observations only). Interpretation, fits and extrapolation are in `../../REPORT.md`. Protocol: `../../PROTOCOL.md`, committed before any measured run (commit `b08afad6c`); curves, ecbench spec and the native Cheon arms frozen in commit `4cbd0ed80` before the first measured run. Every number below is **measured** (operation counts from the arms' ledgers) or a ratio **derived** from them as stated in the column heading.

What ran

item	value
curves	14 frozen prime-order curves, `../../curves.json` (sha256 `8f5f2fd6...9b5b`): 7 sizes (32 to 56 bits) x 2 families (`minus`: `n-1 = d*q`, d about `n^0.129`; `plus`: `n+1 = d*q`, d about `n^0.211`; q prime)
targets	6 planted single-target workloads per curve, `target_seed = 20261022`, ecbench law `uniform_scalar_sha256_v1`; 84 workloads, identical ids in the rho session and the Cheon runs (checked)
arm `rho.negation`	`ecbench run` on `../../ecbench-spec.json` (spec `ECS1hc7a2ce8af299`), session `ECBS1he8d120f81db5` in `../../sessions/rho-20261022/`; 84 runs requested, 84 completed, 84 verified. `ecbench verify --replay 12`: audit OK, 12 of 12 reproduced; `--replay-all`: 84 of 84 reproduced, 0 problems (`../../sessions/rho-20261022.audit.json`, sha256 `e81a637b...fc15`)
arms `cheon.pminus1_comb`, `cheon.pminus1_naive`	`cheon_p224_shape run --family minus`: 84 runs requested, 84 completed, 84 recovered the planted logarithm
arm `cheon.pplus1_comb`	`cheon_p224_shape run --family plus`: 42 runs requested, 42 completed, 42 recovered the planted logarithm
timeouts / failures	none (Table 1b)
binaries	`cheon_p224_shape` sha256 `3c15195b...29bc`; `ecbench` sha256 `0874d63f...5a62f` (built from this worktree on `origin/main` sources; the session's `git_commit` field is null)
host	Intel Xeon @ 2.80 GHz, 4 cores (shared with other jobs, load average 8 to 10 during the run), Linux 6.18.44, ecbench env class `ECBENV2h022d9c040ce5`. Isolation level **LO** (`--cpus none --allow-busy`): operation counts only, wall time is not reported as a result
unit	group operations: one affine addition or doubling = 1 (`ecbench.gae`); every phase charged; identity additions not counted
raw records	`../../runs/cheon-minus.jsonl`, `../../runs/cheon-plus.jsonl`, `../../runs/cheon-all.jsonl` (sha256 `94bc2385...899c`), `../../sessions/rho-20261022/records.jsonl` (sha256 `f3c2bf06...5edc`)
controls	`../../xcheck.txt`: the existing `src/cryptanalysis/cheon_attack.rs` on the 32, 36 and 40-bit `minus` curves (3 targets each) recovers the same logarithm with the same exponentiation count as `cheon.pminus1_naive` (9 of 9 MATCH). `../../replay-certificate.json`: 160 re-executions in one fresh process (every Cheon run up to 48 bits and target 0 at 52 and 56 bits, plus the matched `rho.negation` on the same workloads with the session's seeds), 160 of 160 identical counts and recovered scalars. `../../validator-report.json`: separate program recomputes 45 headline numbers from the raw records, pass. Unit tests (positive and negative controls): `tests/cheon_counted.rs`

Curve identities. Each curve's ICV1 slug and EC1 alias, as computed by ecbench (`registered=false`: not in `docs/curves/registry.json`), are in `../../ecbench-spec.json`'s plan output (`sessions/rho-20261022/plan.json`); the study names them `minus-32` ... `plus-56` by family and size.

Observations

Table 1. Measured operation counts, every run

Unit: group operations (one affine addition or doubling = 1), every phase charged. `S = median ops / sqrt(n)`. `F(n,d)` is the formula `sqrt(n/d)+sqrt(d)` (minus) or `sqrt(n/d)+d` (plus). `ratio to rho` is the median over targets of the paired ratio (same curve, same target). `correct` counts recovered logarithms equal to the planted one.

family	bits	n	d	arm	leaked powers	correct	median ops	min	max	S	ops/F	ratio to rho (median of pairs)	ratio of sums
minus	32	3221255537	16	`rho.negation`	0	6/6	36,521	19,231	72,960	0.643	2.6	1 (reference)	1

family	bits	n	d	arm	leaked powers	correct	median ops	min	max	S	ops/F	ratio to rho (median of pairs)	ratio of sums
minus	32	3221255537	16	cheon.pminus1_comb	2	6/6	56,960	42,502	68,568	1.004	4.0	1.31	1.45
minus	32	3221255537	16	cheon.pminus1_naive	2	6/6	986,396	665,226	1,243,458	17.380	69.5	22.21	25.18
minus	36	51539573731	30	rho.negation	0	6/6	253,101	150,301	522,990	1.115	6.1	1 (reference)	1
minus	36	51539573731	30	cheon.pminus1_comb	2	6/6	159,115	118,228	179,279	0.701	3.8	0.66	0.56
minus	36	51539573731	30	cheon.pminus1_naive	2	6/6	3,377,689	2,347,739	3,885,036	14.878	81.5	14.12	11.79
minus	40	824632004309	28	rho.negation	0	6/6	444,282	71,008	1,447,215	0.489	2.6	1 (reference)	1
minus	40	824632004309	28	cheon.pminus1_comb	2	6/6	818,743	552,699	935,245	0.902	4.8	1.70	1.32
minus	40	824632004309	28	cheon.pminus1_naive	2	6/6	15,237,704	10,234,088	17,428,404	16.780	88.8	31.68	24.64
minus	44	13194139832747	74	rho.negation	0	6/6	3,285,487	2,167,521	4,683,393	0.905	7.8	1 (reference)	1
minus	44	13194139832747	74	cheon.pminus1_comb	2	6/6	2,004,198	1,549,214	2,431,239	0.552	4.7	0.61	0.61
minus	44	13194139832747	74	cheon.pminus1_naive	2	6/6	41,320,964	31,860,597	50,204,230	11.376	97.9	12.58	12.56
minus	48	211106219695673	56	rho.negation	0	6/6	21,834,523	2,843,639	32,193,596	1.503	11.2	1 (reference)	1
minus	48	211106219695673	56	cheon.pminus1_comb	2	6/6	7,285,384	6,051,895	8,837,789	0.501	3.8	0.35	0.36
minus	48	211106219695673	56	cheon.pminus1_naive	2	6/6	165,247,846	137,141,584	200,622,954	11.373	85.1	7.89	8.06
minus	52	3377699780661797	164	rho.negation	0	6/6	61,986,494	39,293,404	120,295,824	1.067	13.7	1 (reference)	1
minus	52	3377699780661797	164	cheon.pminus1_comb	2	6/6	21,471,423	18,487,349	30,042,391	0.369	4.7	0.32	0.33
minus	52	3377699780661797	164	cheon.pminus1_naive	2	6/6	399,345,336	343,790,499	558,913,697	6.871	88.0	5.90	6.17
minus	56	54043195377297631	210	rho.negation	0	6/6	158,861,495	134,220,877	231,679,321	0.683	9.9	1 (reference)	1
minus	56	54043195377297631	210	cheon.pminus1_comb	2	6/6	100,364,872	82,959,549	122,760,558	0.432	6.3	0.59	0.60
minus	56	54043195377297631	210	cheon.pminus1_naive	2	6/6	2,017,940,940	1,667,841,393	2,468,411,403	8.680	125.8	11.94	12.02
plus	32	3221175689	90	rho.negation	0	6/6	51,436	27,244	93,796	0.906	8.5	1 (reference)	1
plus	32	3221175689	90	cheon.pplus1_comb	180	6/6	95,294	58,721	113,909	1.679	15.7	1.82	1.59
plus	36	51539736719	240	rho.negation	0	6/6	187,614	114,238	346,591	0.826	12.6	1 (reference)	1
plus	36	51539736719	240	cheon.pplus1_comb	480	6/6	222,988	150,802	248,849	0.982	15.0	1.08	1.00
plus	40	824633358973	398	rho.negation	0	6/6	1,087,652	214,874	2,075,309	1.198	23.7	1 (reference)	1
plus	40	824633358973	398	cheon.pplus1_comb	796	6/6	704,630	388,117	941,678	0.776	15.3	0.70	0.55
plus	44	13194137032873	574	rho.negation	0	6/6	4,150,295	1,490,804	4,632,503	1.143	27.3	1 (reference)	1
plus	44	13194137032873	574	cheon.pplus1_comb	1148	6/6	2,419,814	1,509,216	2,845,511	0.666	15.9	0.63	0.66
plus	48	211106244944429	990	rho.negation	0	6/6	10,502,984	504,689	19,888,818	0.723	22.7	1 (reference)	1
plus	48	211106244944429	990	cheon.pplus1_comb	1980	6/6	5,077,816	3,626,295	9,339,109	0.349	11.0	0.63	0.59
plus	52	3377699707176467	1836	rho.negation	0	6/6	64,022,410	39,514,105	119,465,498	1.102	47.1	1 (reference)	1
plus	52	3377699707176467	1836	cheon.pplus1_comb	3672	6/6	22,508,339	12,957,180	35,433,734	0.387	16.6	0.33	0.34
plus	56	54043195250336543	2016	rho.negation	0	6/6	429,577,620	84,161,122	519,640,803	1.848	82.9	1 (reference)	1
plus	56	54043195250336543	2016	cheon.pplus1_comb	4032	6/6	98,342,762	49,002,962	131,761,087	0.423	19.0	0.21	0.24

Table 1b. Failures, wrong answers and timeouts (preserved)

None: every recorded run returned the planted logarithm.

Table 2. Unpriced work and the pricing band (AGENTS.md 5a)

Unpriced counters per arm (median over targets), then the Cheon/rho cost ratio when every unpriced operation is priced at `w` group-addition equivalents, the same `w` for both arms. Order is **resolved** only if the ratio is on the same side of 1 at w = 0, 0.1 and 1.

family	bits	arm	unpriced counters (median)	ratio w=0	ratio w=0.1	ratio w=1	order
minus	32	`rho.negation`	canonicalisations_uncharged=36,120	1.00	1.00	1.00	reference
minus	32	`cheon.pminus1_comb`	scalar_mulmod_uncharged=22,220, table_insert_uncharged=14,195, table_lookup_uncharged=8,028	1.31	1.28	1.15	dearer than rho at all weights
minus	32	`cheon.pminus1_naive`	scalar_mulmod_uncharged=22,220, table_insert_uncharged=14,195, table_lookup_uncharged=8,028	22.21	20.30	11.66	dearer than rho at all weights
minus	36	`rho.negation`	canonicalisations_uncharged=252,686	1.00	1.00	1.00	reference
minus	36	`cheon.pminus1_comb`	scalar_mulmod_uncharged=67,098, table_insert_uncharged=41,455, table_lookup_uncharged=25,646	0.66	0.65	0.61	cheaper than rho at all weights
minus	36	`cheon.pminus1_naive`	scalar_mulmod_uncharged=67,098, table_insert_uncharged=41,455, table_lookup_uncharged=25,646	14.12	12.89	7.35	dearer than rho at all weights
minus	40	`rho.negation`	canonicalisations_uncharged=443,855	1.00	1.00	1.00	reference
minus	40	`cheon.pminus1_comb`	scalar_mulmod_uncharged=270,458, table_insert_uncharged=171,620, table_lookup_uncharged=98,840	1.70	1.65	1.42	dearer than rho at all weights
minus	40	`cheon.pminus1_naive`	scalar_mulmod_uncharged=270,458, table_insert_uncharged=171,620, table_lookup_uncharged=98,840	31.68	28.91	16.41	dearer than rho at all weights
minus	44	`rho.negation`	canonicalisations_uncharged=3,285,050	1.00	1.00	1.00	reference
minus	44	`cheon.pminus1_comb`	scalar_mulmod_uncharged=662,880, table_insert_uncharged=422,264, table_lookup_uncharged=240,618	0.61	0.59	0.51	cheaper than rho at all weights
minus	44	`cheon.pminus1_naive`	scalar_mulmod_uncharged=662,880, table_insert_uncharged=422,264, table_lookup_uncharged=240,618	12.58	11.48	6.49	dearer than rho at all weights
minus	48	`rho.negation`	canonicalisations_uncharged=21,834,071	1.00	1.00	1.00	reference
minus	48	`cheon.pminus1_comb`	scalar_mulmod_uncharged=2,418,184, table_insert_uncharged=1,941,594, table_lookup_uncharged=476,592	0.35	0.34	0.29	cheaper than rho at all weights
minus	48	`cheon.pminus1_naive`	scalar_mulmod_uncharged=2,418,184, table_insert_uncharged=1,941,594, table_lookup_uncharged=476,592	7.89	7.20	4.06	dearer than rho at all weights
minus	52	`rho.negation`	canonicalisations_uncharged=61,986,040	1.00	1.00	1.00	reference
minus	52	`cheon.pminus1_comb`	scalar_mulmod_uncharged=5,372,230, table_insert_uncharged=4,538,265, table_lookup_uncharged=833,968	0.32	0.30	0.24	cheaper than rho at all weights
minus	52	`cheon.pminus1_naive`	scalar_mulmod_uncharged=5,372,230, table_insert_uncharged=4,538,265, table_lookup_uncharged=833,968	5.90	5.38	3.03	dearer than rho at all weights
minus	56	`rho.negation`	canonicalisations_uncharged=158,861,022	1.00	1.00	1.00	reference
minus	56	`cheon.pminus1_comb`	scalar_mulmod_uncharged=25,119,560, table_insert_uncharged=16,042,102, table_lookup_uncharged=9,077,460	0.59	0.57	0.45	cheaper than rho at all weights
minus	56	`cheon.pminus1_naive`	scalar_mulmod_uncharged=25,119,560, table_insert_uncharged=16,042,102, table_lookup_uncharged=9,077,460	11.94	10.89	6.12	dearer than rho at all weights
plus	32	`rho.negation`	canonicalisations_uncharged=51,046	1.00	1.00	1.00	reference
plus	32	`cheon.pplus1_comb`	fp2_mul_uncharged=9,980, scalar_mulmod_uncharged=1,448, table_insert_uncharged=5,993, table_lookup_uncharged=3,989	1.82	1.69	1.12	dearer than rho at all weights
plus	36	`rho.negation`	canonicalisations_uncharged=187,212	1.00	1.00	1.00	reference
plus	36	`cheon.pplus1_comb`	fp2_mul_uncharged=23,606, scalar_mulmod_uncharged=3,848, table_insert_uncharged=14,671, table_lookup_uncharged=8,937	1.08	1.00	0.66	unresolved
plus	40	`rho.negation`	canonicalisations_uncharged=1,087,237	1.00	1.00	1.00	reference

family	bits	arm	unpriced counters (median)	ratio w=0	ratio w=0.1	ratio w=1	order
plus	40	`cheon.pplus1_comb`	fp2_mul_uncharged=70,532, scalar_mulmod_uncharged=6,376, table_insert_uncharged=45,539, table_lookup_uncharged=24,994	0.70	0.65	0.42	cheaper than rho at all weights
plus	44	`rho.negation`	canonicalisations_uncharged=4,149,856	1.00	1.00	1.00	reference
plus	44	`cheon.pplus1_comb`	fp2_mul_uncharged=249,926, scalar_mulmod_uncharged=9,192, table_insert_uncharged=151,637, table_lookup_uncharged=98,290	0.63	0.59	0.38	cheaper than rho at all weights
plus	48	`rho.negation`	canonicalisations_uncharged=10,502,537	1.00	1.00	1.00	reference
plus	48	`cheon.pplus1_comb`	fp2_mul_uncharged=608,362, scalar_mulmod_uncharged=15,848, table_insert_uncharged=461,810, table_lookup_uncharged=146,554	0.63	0.59	0.39	cheaper than rho at all weights
plus	52	`rho.negation`	canonicalisations_uncharged=64,021,948	1.00	1.00	1.00	reference
plus	52	`cheon.pplus1_comb`	fp2_mul_uncharged=1,981,204, scalar_mulmod_uncharged=29,384, table_insert_uncharged=1,356,401, table_lookup_uncharged=624,805	0.33	0.30	0.20	cheaper than rho at all weights
plus	56	`rho.negation`	canonicalisations_uncharged=429,577,158	1.00	1.00	1.00	reference
plus	56	`cheon.pplus1_comb`	fp2_mul_uncharged=8,316,886, scalar_mulmod_uncharged=32,264, table_insert_uncharged=5,177,607, table_lookup_uncharged=3,139,280	0.21	0.20	0.12	cheaper than rho at all weights

Table 6. Where the operations go (median share of total, per phase)

`cheon.pminus1_comb` (minus). Shares in percent.

bits	p-1:precompute	p-1:step1:baby	p-1:step1:giant	p-1:step2:baby	p-1:step2:giant	p-1:step2:precompute	p-1:verify
32	21.6	49.8	28.1	0.1	0.0	0.3	0.1
36	15.5	52.1	32.2	0.0	0.0	0.2	0.0
40	1.0	63.3	35.6	0.0	0.0	0.0	0.0
44	0.8	63.2	35.9	0.0	0.0	0.0	0.0
48	0.5	80.4	19.2	0.0	0.0	0.0	0.0
52	0.1	84.5	15.4	0.0	0.0	0.0	0.0
56	0.0	64.2	35.8	0.0	0.0	0.0	0.0

`cheon.pplus1_comb` (plus). Shares in percent.

bits	p+1:poly	p+1:step1:baby	p+1:step1:giant	p+1:step1:precompute	p+1:step2:baby	p+1:step2:giant	p+1:step2:precompute	p+1:step2:shift	p+1:verify
32	12.8	25.1	41.7	19.4	0.1	0.1	0.4	0.3	0.0
36	16.6	26.3	40.0	16.6	0.1	0.1	0.2	0.1	0.0
40	9.7	38.8	49.4	1.8	0.0	0.1	0.1	0.1	0.0
44	4.5	37.9	56.5	1.0	0.0	0.0	0.0	0.0	0.0
48	4.1	55.5	39.4	1.0	0.0	0.0	0.0	0.0	0.0
52	2.0	52.2	45.7	0.1	0.0	0.0	0.0	0.0	0.0
56	0.5	42.2	57.3	0.1	0.0	0.0	0.0	0.0	0.0

Graphs

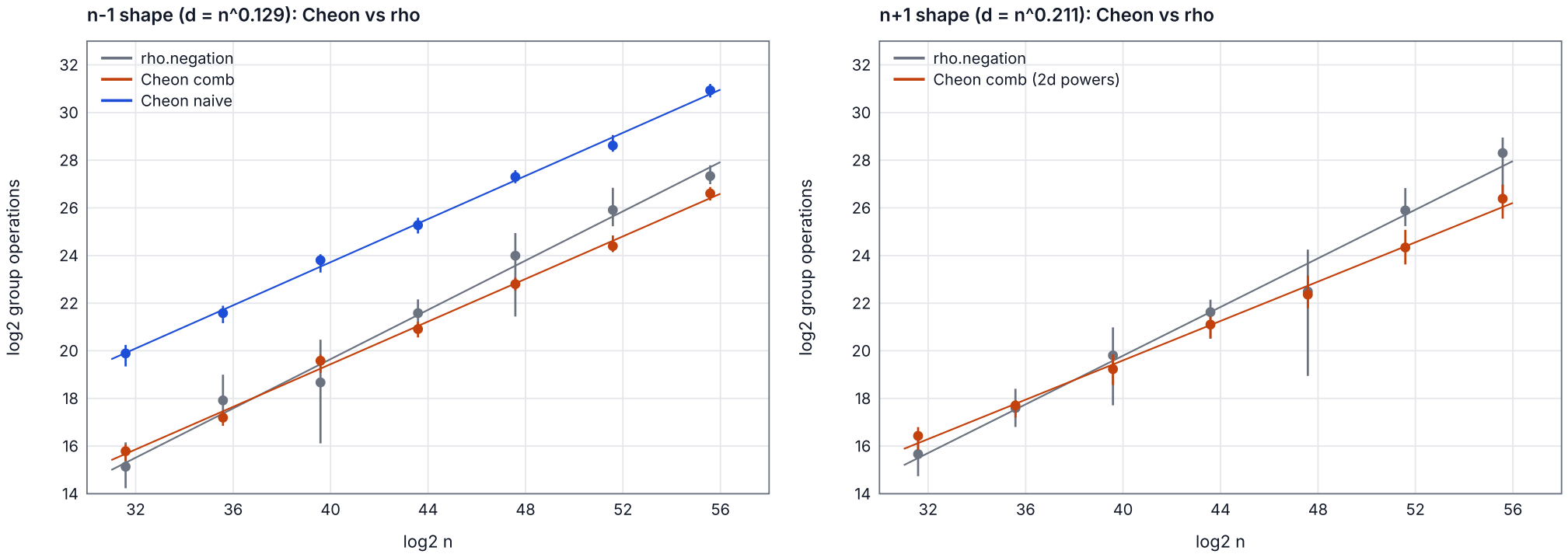


Fig. 1. MEASURED log2 operations per size (dots = mean over verified targets, bars = min..max) with fit A over the measured range. Source: results.json.

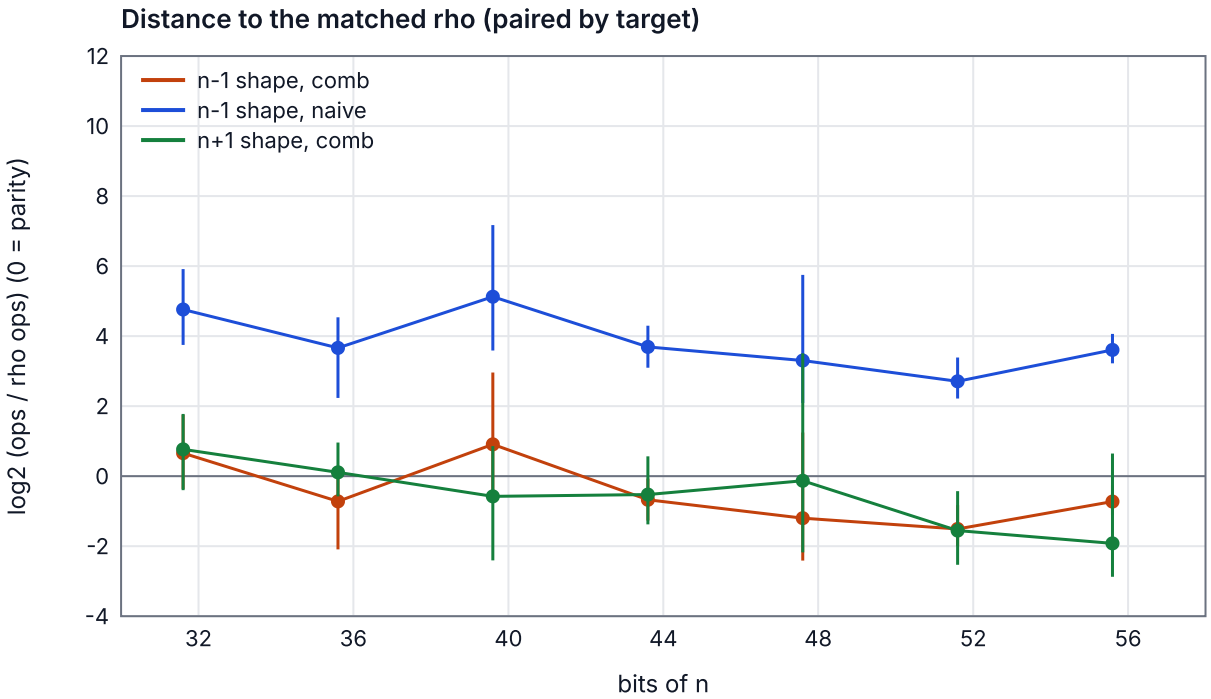


Fig. 2. log2 of the Cheon/rho operation ratio per size (dots = mean, bars = min..max over paired targets). Above 0 = Cheon dearer. Source: runs.jsonl and the ecbench rho session.

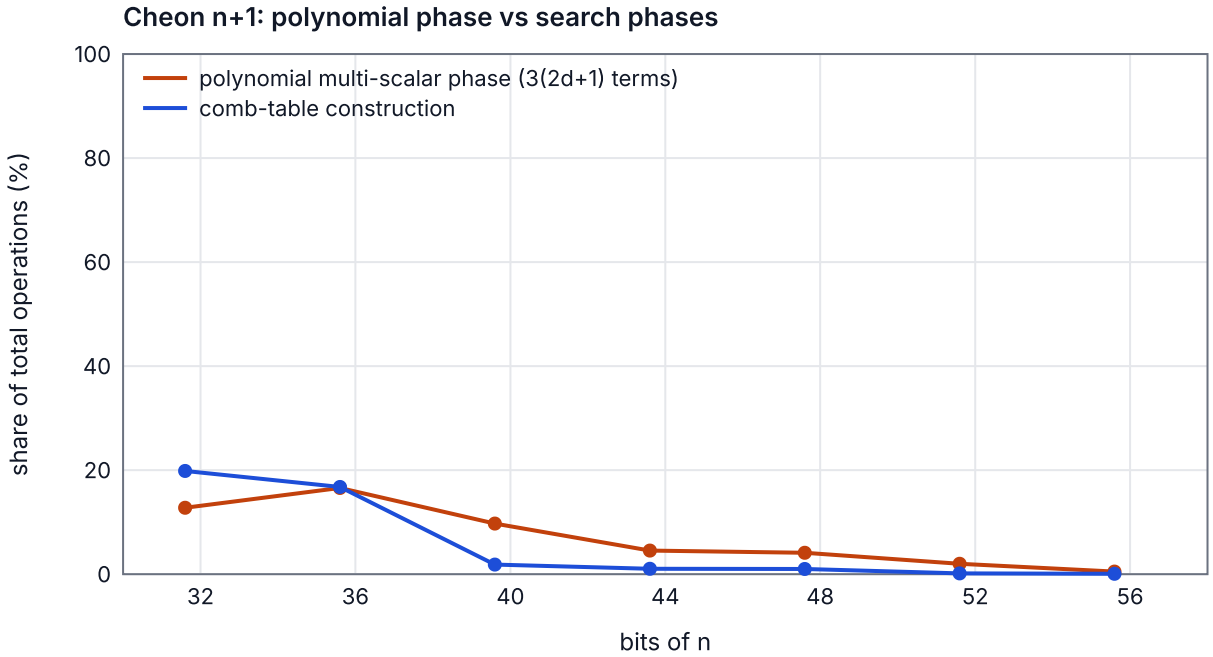


Fig. 3. Median share of the n+1 variant's operations by phase. Source: runs.jsonl phase columns.

Graph data: `../../results/results.json`; sources: `../../runs/*.jsonl` and the rho session records. Figures are drawn by `examples/cheon_p224_shape_analyse.rs` from recorded values only. Fig. 4 (extrapolation to 224 bits) is a calculation and is in `../../REPORT.md`.